

ANTIBACTERIAL POTENTIAL OF BENZIMIDAZOLE CONTAINING HYBRID DERIVATIVES

Suvaiv¹, Kuldeep Singh^{1,✉}, Praveen Kumar², Syed Misbahul Hasan¹, Shom
Prakash Kushwaha¹, Arun Kumar¹, Syed Mehdi Hasan Zaidi¹
and Shubham Kumar Maurya¹

¹Faculty of Pharmacy, Integral University Lucknow, 226026, UP, India

²Faculty of Pharmacy, Uttar Pradesh University of Medical Sciences, Saifai, Etawah, 206130, UP, India

✉Corresponding Author: kulintegral@gmail.com

ABSTRACT

A wide range of bacterial infections are mitigated with antibiotics, however, repeated infections produced by importunate bacteria are obstructing effective treatment. Furthermore, the resistance of bacteria to the existing antibiotics is a significant and vexing problem of the day. The hybrid pharmacophore structural modification concept is now widely used to develop new medications with synergistic activity or increased medication efficacy. This review looks at the benzimidazole hybrids that have evolved in the last two decades and have probable antibacterial action. Structure-activity relationship and study of molecular interaction are also examined to locate the course for the intent and expansion of benzimidazole hybrids by means of good pharmacokinetic and pharmacodynamic profiles. Based on the findings, it can be inferred that such hybrids that have substitution on both ring benzimidazole, as well as other heterocyclic rings, have better antibacterial potential rather than the hybrids that have substitution only on a single ring. Findings also indicate the crucial positions for substitution to have maximum antibacterial potential. The antibacterial activity not only depends on the substitution of rings but also depends on the attachment of the linker group between two rings. Resulting, it is thought worthwhile to synthesize novel benzimidazole hybrids having appropriate linkers and substituents with the goal of having potent bactericidal activity.

Keywords: Benzimidazole, Antibacterial, Hybrid derivatives, SAR, Antimicrobial resistance, Heterocyclic ring.

RASAYAN J. Chem., Vol. 16, No.3, 2023

INTRODUCTION

A variety of infectious diseases caused by various bacterial strains are found to be resistant to existing antibiotics.^{1,2} The antibacterial potential of existing antibiotics decreases due to the unfortunate use of antibiotics.³⁻⁶ Antimicrobial resistance is becoming a leading cause of death in low and middle-income countries.⁷ The seriousness of the problem can be estimated through current data of deaths per year associated with antimicrobial-resistance is given in GRAM's (The Global Research on Antimicrobial Resistance) report.⁸ Resulting, drug-challenging bacteria presented a severe risk and it is critical to develop novel antibacterial drug candidates that are both effective and unique.⁹ Several benzimidazole derivatives have been produced in recent years and shown to have a range of pharmacological properties including antibacterial¹⁰ and a few synthetic benzimidazole derivatives were also shown to have antihelmintic properties.¹¹ N-azopridinium salts and various betaines were also discovered to have antiprotozoal activity. From the available literature, it can be deduced that the benzimidazole molecule has a variety of biological actions, including antihistaminic, anticancer, cytotoxic¹², antitubercular¹³, antifungal¹⁴, antiviral¹⁴, and anti-inflammatory properties.¹⁶ Because of their clinical uses and biological activities like antimicrobial¹⁷, anticonvulsant, anti-depressive, antihypertensive, analgesic, enzyme inhibition, and anti-diabetic effects¹⁸ medicinal chemistry is interested in their derivatives. The hybridization of various pharmacophores into a single molecule (Fig.-1) is a valuable method in the discovery of novel pharmacological components.¹⁹⁻²⁷

This review wraps up the latest expansion of benzimidazole hybrids as potent antibacterial candidates casing papers published in the last two decades. SARs (structure-activity relationships) are examined in order to offer guidance toward the reasonable development of efficacious candidates.

Reported Antibacterial Activity of Some Benzimidazole Derivatives

Phenyl-substituted benzimidazole derivatives having electronegative groups linked to pyrazole (at C-5 position) were found to have a superior antibacterial activity to those with an electron-releasing group. Furthermore, the unsubstituted compounds had fair activity.

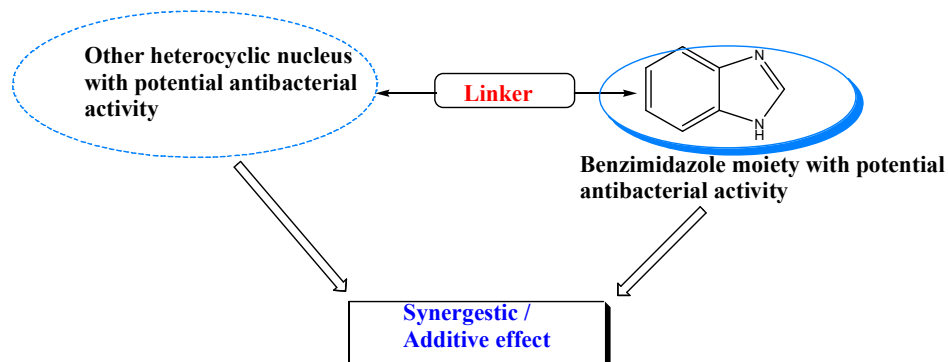


Fig.-1: Hybridization of Benzimidazole with other Antibacterial Heterocycles

Compound (1) outperformed the other compounds examined. As a result, this chemical can be used as a lead molecule in the expansion of a clinically relevant antibacterial drug.²⁸ SAR of title compounds (1) shown in Fig.-2.

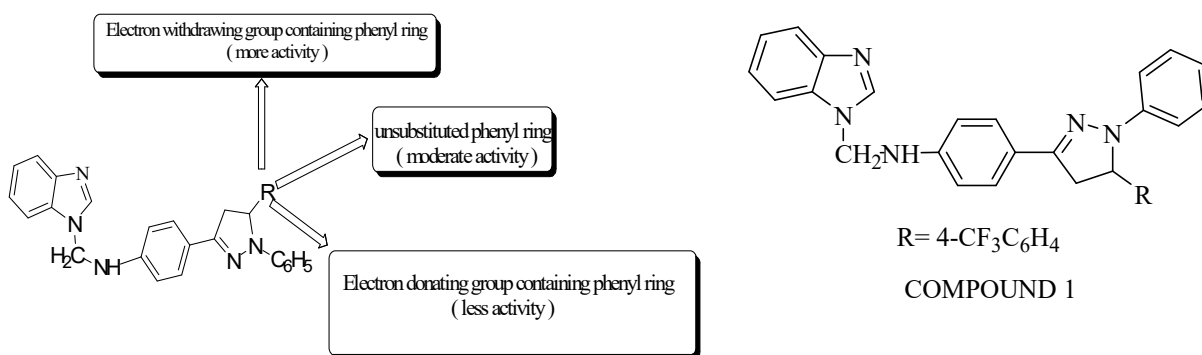


Fig.-2: Schematic Representation of SAR of Title Compounds (1)

A compact system containing benzimidazole and oxadiazole groups was created with the goal of developing molecules with interesting pharmacological characteristics. Compound 2, which has electron-withdrawing para-nitro substituents, was shown the best antibacterial activity outperforming chloramphenicol.²⁹

A SAR study revealed a possible link between the structures of this compound class and antibacterial activity: the para position of the phenyl ring is more crucial for the substitution with electronegative groups for more effective proposed activity. Other research groups have backed up the idea that an inductive electron-withdrawing group can boost antibacterial activity.³⁰ Desai and colleagues also created two further benzimidazole series: pyrazoline-bearing derivatives (Fig.-3) and benzimidazoles containing cyanopyridine and 4-thiazolidinone (Fig.-4).³¹ Compound 3 had the best antibacterial activity and compound 4 was discovered to have greater action against *Pseudomonas aeruginosa* and *Streptococcus pyogenes* (25 µg/mL), as well as high inhibitory activity (12.5 µg/mL) against *E. coli*, than reference chloramphenicol (50 µg/mL).³² The antibacterial performance of a novel division of antibacterial drugs (Triaryl benzimidazoles) evaluated against multidrug-resistant bacterial strain.

Due to a change in the form of the molecules, molecular modeling investigations reveal that the newly created category cannot fit through the minor groove of DNA. Compounds 5, 6, and 7 were discovered to be potent among all, with MICs between 0.5 to 4 µg/mL against multi-drug resistant *Staphylococci* and *Enterococci*. Compound 6, against multidrug-resistant bacterial strain, was found to have modest activity with a minimum inhibitory concentration between 16 to 32 µg/mL.³³

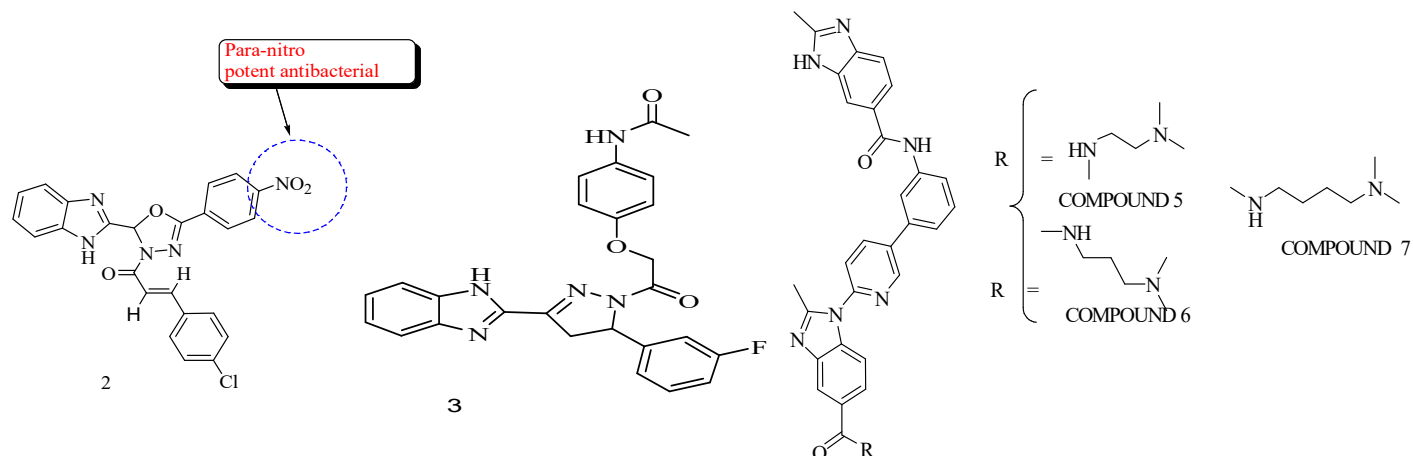


Fig.-3: Pyrazoline-bearing derivatives of benzimidazole

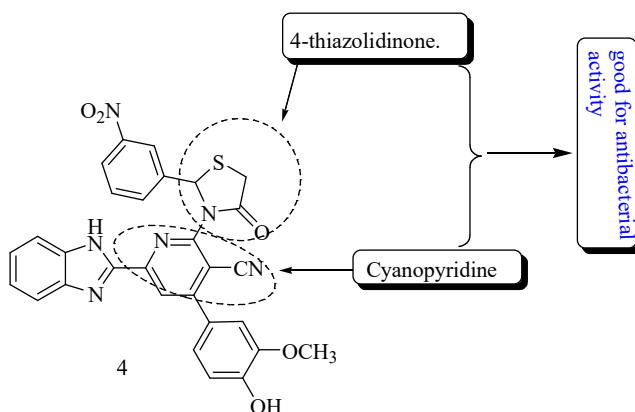


Fig.-4: Cyanopyridine and 4-thiazolidinone derivatives of benzimidazole

Different 2-phenyl benzimidazole derivatives have been described as powerful antibacterial with fair to moderate efficacy by a group of researchers (Turkey), against *Candida albicans* and certain bacterial types. Substituents with a 2-phenyl ring at meta and para have been generated, Compounds 8 ($R = \text{NH}_2$) and 9, which have antibacterial activity comparable to that of ampicillin and against *Pseudomonas aeruginosa*.^{34, 35} Structural constraints around the benzimidazole nucleus requires for antimicrobial action is described in Fig.-5.

Antibacterial Activity of Benzimidazole and Other Heterocyclic Hybrid Derivatives:

Coumarin–benzimidazole Hybrids as Potent Antibacterial Agents

The antibacterial potential of eleven proposed novel conjugates has been evaluated against a panel of the pathogenic bacterial strain. Amongst all, hybrids 10 and 11 (Fig.-6) efficiently showed bactericidal potential against *Bacillus subtilis* with MIC=0.95 and 6.25 $\mu\text{g/mL}$, correspondingly, while standard drug Ampicillin had MIC=25 $\mu\text{g/mL}$. Compounds 11, 12, 13, and 14 showed substantial suppression of *Proteus vulgaris* (ATCC 29905) at MIC 1.56, 3.12, 1.56, and 1.56 $\mu\text{g/mL}$, in that order, compounds having a halogen atom have antibacterial activity, according to molecular structure studies. Compound 10, -Cl atom (at para position) was discovered to have the most wide-ranging- antibacterial activity. Surprisingly, compounds having a -Cl atom (Meta or Ortho position) (14) have no bactericidal action.³⁶

Antibacterial Potential of Novel Benzimidazole–Oxadiazole Hybrid Compounds

Antibacterial activity of 20, new benzimidazole–oxadiazole hybrid molecules was recorded against tested bacterial strains, and MICs were compared with ciprofloxacin. Among all, compounds 15 and 16 (Fig.-7) demonstrated the most potent antibacterial agents.

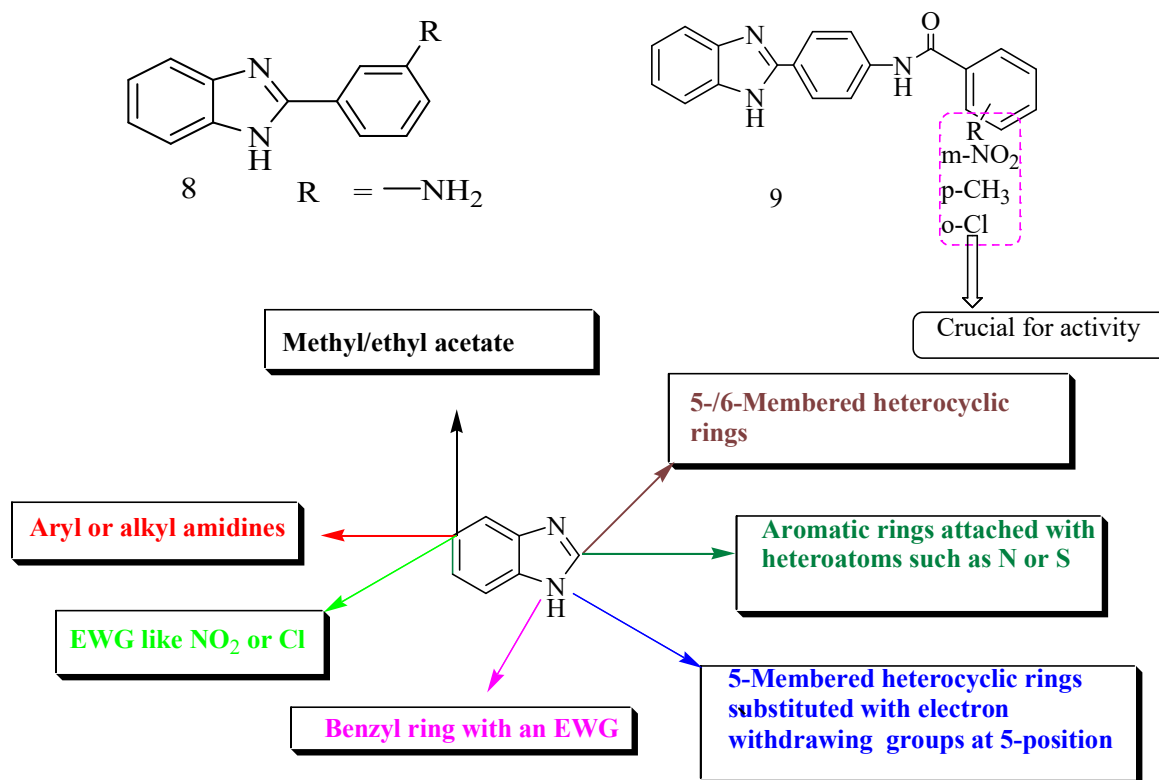


Fig.-5: Antimicrobial Action Requires Structural Constraints Around the Benzimidazole Nucleus

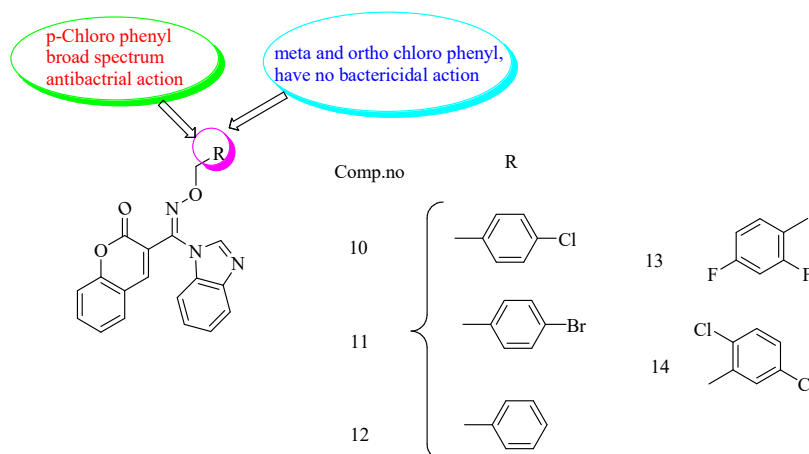


Fig.-6: Coumarin-benzimidazole Hybrids

Compound 15, which had an oxadiazole nucleus with a p-chlorophenyl group at C3, and a 2-chloro-6-fluorophenyl at C2 of the benzimidazole ring, with MIC (1.6 µg/mL against *E. faecalis*, 3.125 µg/mL against *S. aureus*, 6.25 µg/mL). Compound 16, which had a 3-chloro-2-fluorophenyl instead of the 2-chloro group (compound 15) has gram-positive activity (MIC = 1.6 µg/mL). Findings indicate that the superior activity could be recognized due to the existence of halogen atoms; chiefly Fluorine atom could convey potentiality.³⁷

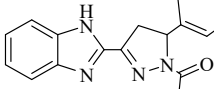
Antibacterial activity of Benzimidazole-Pyrazole and Pyridine hybrids

The pyridine and pyrazole moiety which can block a series of enzymes and receptors is generally used in pharmaceutical research.³⁸ To sustain their pyrimidine pools, bacteria rely on de novo pyrimidine

The presence of chloro-fluoro may explain the increased activity

15

3-OH $\geq$ 3-H
excellent activity
against *P. aeruginosa*.



17a-n

R = Different substituents

R = 17a (-H)	R = 17i (4-OCH ₃)
R = 17b (2-Cl)	R = 17j (4-Cl)
R = 17c (2-OH)	R = 17k (4-Br)
R = 17d (3-OH),	R = 17l (4-F)
R = 17e (3-NO ₂)	R = 17m (4-NO ₂)
R = 17f (4-OH, 3-OCH ₃)	R = 17n, 4-N(CH ₃) ₂
R = 17g (4-OH),	
R = 17h (4-CH ₃),	

Fig.-8: Benzimidazole-Pyrazole and Pyridine hybrids

Molecules containing benzimidazole, benzoxazole, and benzothiazole nucleus in their structure have a broad assortment of biological features such as antiviral⁴², anticancer⁴³, and antimicrobial⁴⁴ actions have been demonstrated for these heterocycles. Pyrazoles are used as essential intermediates, and important precursors in the synthesis of benzofuran/coumarin systems and benzimidazole, benzoxazole, and benzothiazole derivatives respectively, although antibacterial properties of different pyrazole derivatives have been described in the literature.⁴⁵ Many bacteria have been reported to be inhibited by pyrazoles and substituted pyrazoles.⁴⁶ A group of researchers evaluated the *in vitro* antibacterial potential (serial dilution method) of 12 newly synthesized benzimidazoles, benzothiazoles, and benzoxazoles derivatives (Fig.-9). The antibacterial activity of derivatives 18a, 18c, 18d, 18g, 18j, and 18k was good. 18b, 18f, and 18i demonstrated weak to moderate activity as evidenced by their MIC values. 18e and 18h were shown to be the least potential against both tested bacterial strains, among these compounds. The new 2-

[substituted-1H-pyrazol- 4-yl] has a structure–activity connection. In contrast to the benzothiazole moiety (18e and 18f) in target compounds, novel compounds comprising benzimidazole (18b, 18c, 18g, 18h, 18i, 18j, and 18k) and benzoxazole (18a and 18d) demonstrated broad-spectrum antibacterial profiles against tested species. No effect of electron-donating and withdrawing groups in target compounds (18a–18k) is observed on the bactericidal action against the tested bacterial strain.⁴⁷

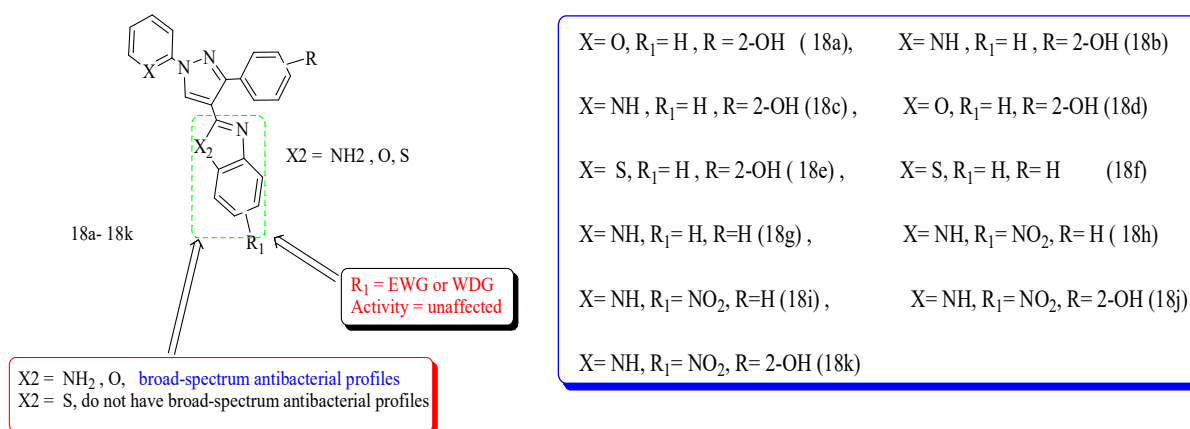
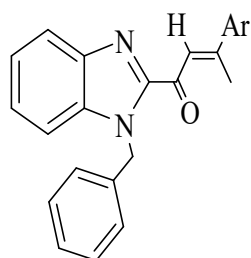


Fig.-9: Benzothiazoles, Benzoxazoles, and Benzimidazole hybrids

Novel 1, 3, 5-trisubstituted Pyrazoline Hybridized with Benzimidazole Analogues as Antibacterial Agents

The pyrazoline motif has to be found in a variety of physiologically active compounds. Antimicrobial^{48,49} anticancer^{50,51} anti-inflammatory^{52,53} and monoamine oxidase inhibitory^{54,55} activities were demonstrated in some members of this class. A new series of antibacterial drugs known as thiazolyl-pyrazoline⁵⁶ has just been discovered. Hybridization of pyrazoline and N-benzyl benzimidazole fragments may serve as a springboard for the discovery of novel antibacterial agents.¹⁵ New 1,3,5-trisubstituted pyrazoline analogs hybridized with benzimidazole derivatives were developed (Fig.-10) and recorded for bactericidal potential against selected pathogenic strains. Compound 28 (minimum inhibitory concentration= 64µg/mL) has been found to be the most potent. MICs were compared with ciprofloxacin (MIC = 6.25–12.5µg/mL). SAR indicates that the electronegative groups on the benzene nucleus exhibited good antimicrobial activity than compounds with positive inductive effects. As seen here, compound 28, MIC of 64µg/mL, in comparison with MIC of compound 24, which was reported to be 256 and 512µg/mL, against *Staphylococcus aureus* and *Escherichia coli*, correspondingly.⁵⁷



19-23

Compound 19,24,29	Ar = Phenyl
Compound 20,25,30	Ar = 4-Methylphenyl
Compound 21,26,31	Ar = 4-Chlorophenyl
Compound 22,27,32	Ar = 4-Bromophenyl
Compound 23,28,33	Ar 4-Fluorophenyl

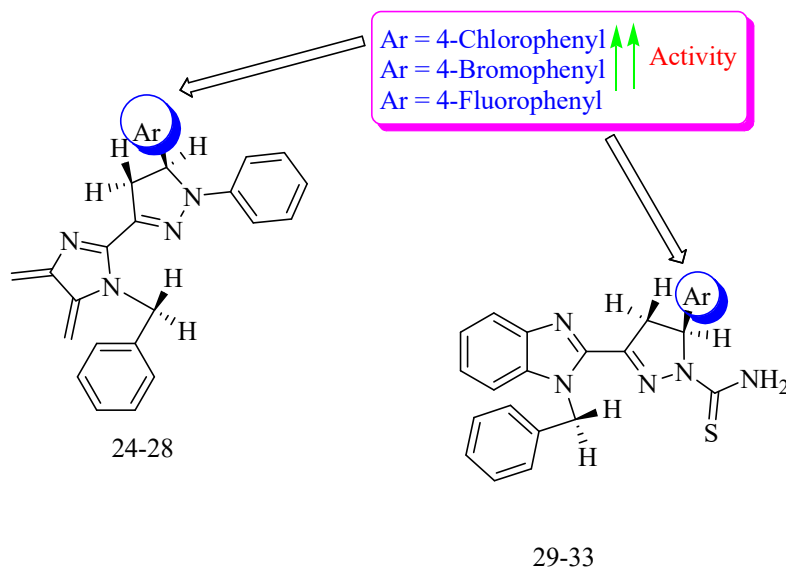


Fig.-10: Hybridization of Pyrazoline with Benzimidazole

Novel Benzimidazole- Pyrimidine/Pyrazole/Isoxazole Hybrid Derivatives as Antibacterial Agents

13 novel benzimidazole hybrid with pyrimidine/pyrazole/isoxazole have been developed (Fig.-11) and their *in vitro* antibacterial potential was recorded (agar streak dilution method). Amongst all 13 compounds, 34 and 35 were found to be more potent for their antibacterial potential, as shown by their minimum inhibitory concentrations. The MIC values of compounds 34 and 35 were similar to the Ciprofloxacin against *Staphylococcus epidermidis* (MIC: 1.95 µg/mL), *S. aureus* (MIC: 7.81 µg/mL), and *E. coli* (MIC: 7.81 µg/mL).⁵⁸

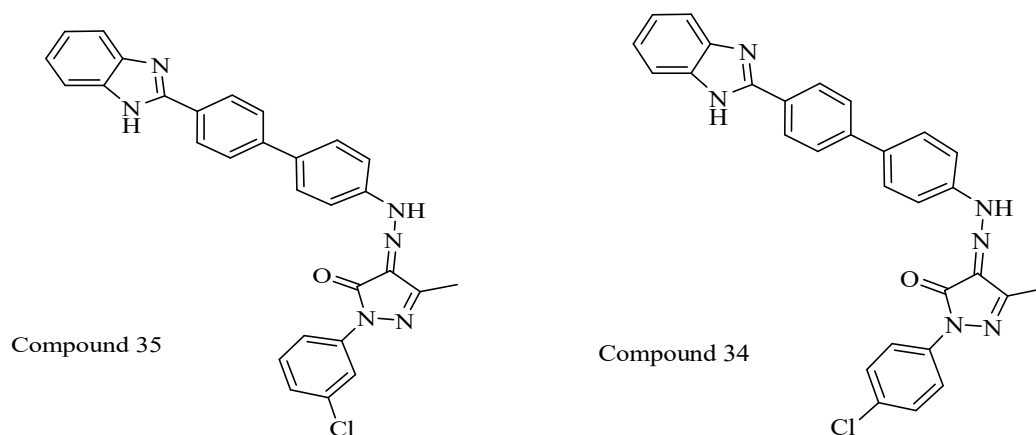


Fig.-11: Benzimidazole- Pyrimidine/Pyrazole/Isoxazole Hybrid Derivatives

Antibacterial Potential of Novel Pyrazoles and Pyrazolyl Thiazoles Containing Benzimidazole Derivatives

21 Novel hybrid derivatives were designed and synthesized (Fig.-12) and the antibacterial potential of the compounds 4, 5, and 7 were estimated *in vitro* against tested strains. When compared to the typical antibiotic Chloramphenicol, compound 5d demonstrated outstanding effectiveness against *P. aeruginosa*. The activity of (bis and tris derivatives, 37 and 38) was found to be considerably higher than that of (mono derivative, 36). This could be owing to the substitution of EWGs in bis heterocyclic compounds (NO₂, F, Cl, and Br). Derivatives with these (electron withdrawing) groups (37d, 37e, 37f, 37g, 38d, 38e, 38f, and 38g) have superior activity in comparison with those having donating groups

(OCH₃ and CH₃), (38d, 38e, 38f, and 38g). Indeed, when compared to the conventional antibiotic Chloramphenicol, compound 37f demonstrated outstanding effectiveness against *P. aeruginosa*. This could be because the aromatic ring contains more electronegative chlorine.⁵⁹

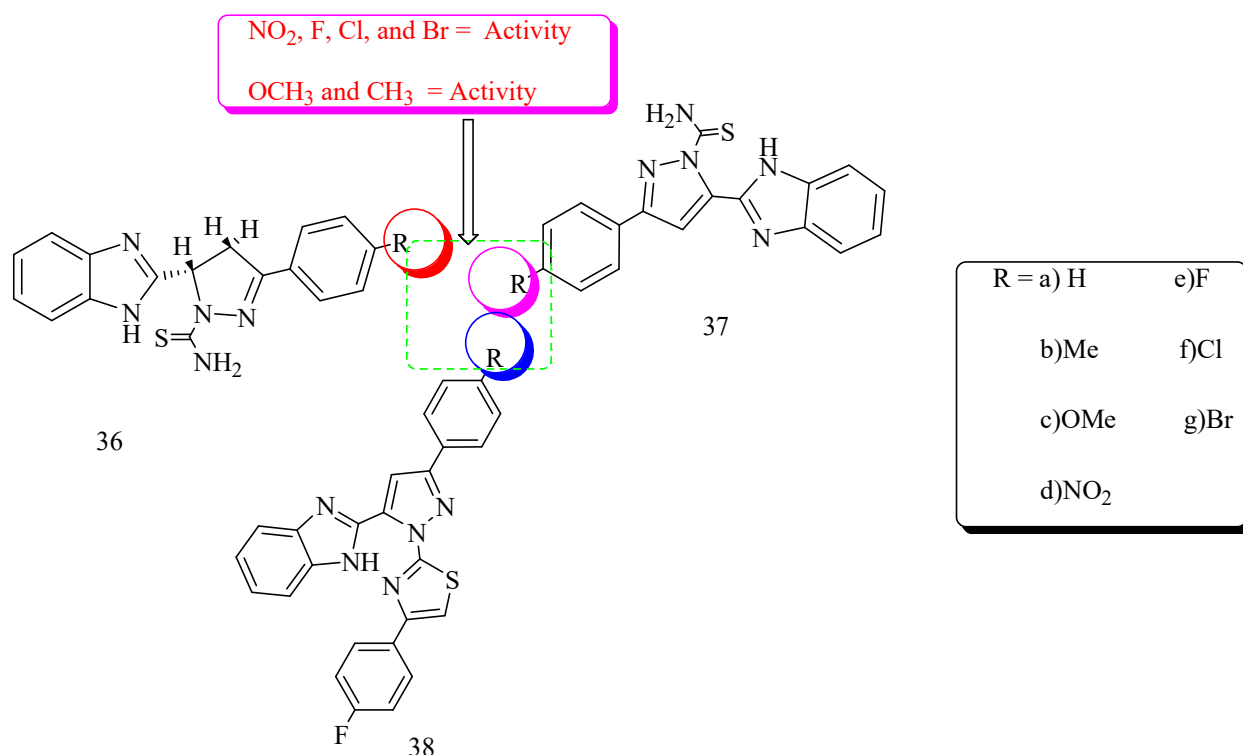


Fig.-12: Pyrazoles and Pyrazolyl Thiazoles Containing Benzimidazole Derivatives

Pyrazolyl-Benzimidazole Hybrids as Antibacterial Agents

The antibacterial potential of synthesized compounds (39, 40, and L, Fig.-13) and metal complexes was recorded against proposed bacterial strains. Findings were matched to those of chloramphenicol at various concentrations as a standard. When compared to 39 and 40, the 39 complexes were showing extreme activity whereas; mononuclear coordination complexes are more active than the free ligand (L).⁶⁰

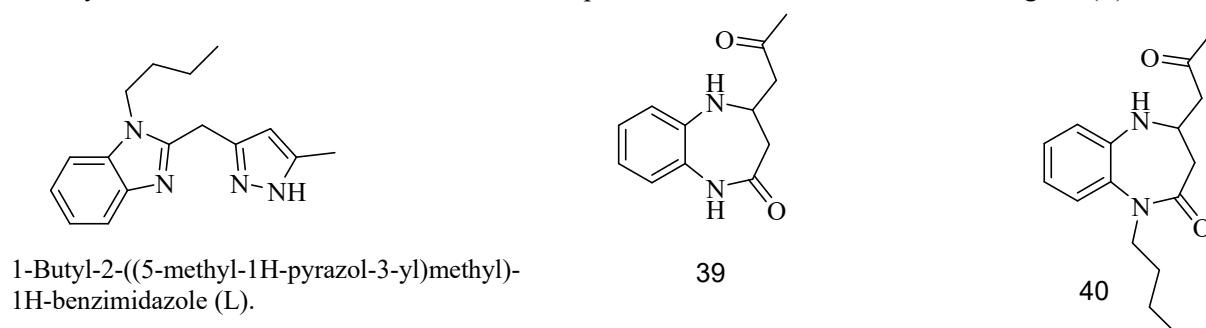


Fig.-13: Pyrazolyl-Benzimidazole Hybrids

Antibacterial Activities of New Benzimidazole-Pyrazole-4-Carbonitriles-Methanone Substituted Derivatives

Antimicrobial potential of new synthesized library of benzimidazole-pyrazole-4-carbonitriles-methanone substituted derivatives (Fig.-14) has been estimated (diffusion plate method) against selected bacterial strains. By comparing the MIC values of synthesized compounds with Amoxicillin, compounds (41a) and compound 44 were shown to be most potent against *S. typhimurium*. Derivatives (42b) and (43b) was excellent potency against *S. aureus*, with an 18 mm inhibition zone.⁶¹

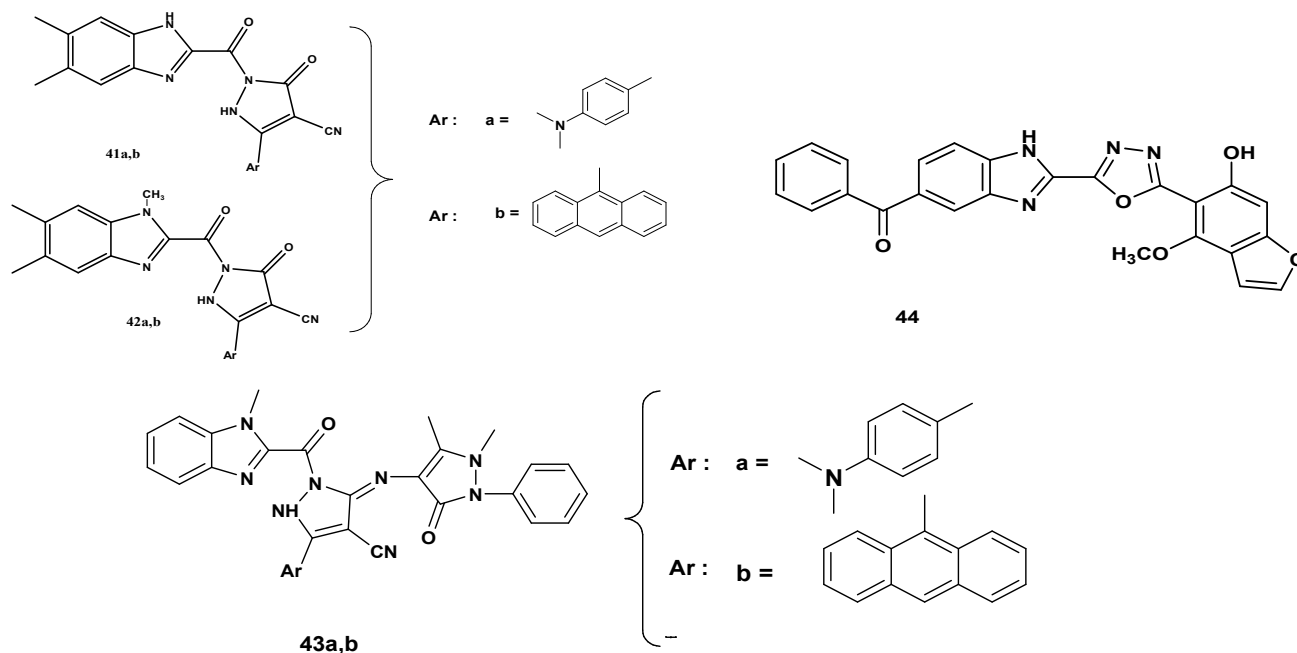


Fig.-14: Benzimidazole-Pyrazole-4-Carbonitriles-Methanone Hybrid Derivatives

Antibacterial Drugs Based on Novel Benzimidazole1, 2, 3 triazole-sulfonamide Hybrids

1, 2, and 3-triazoles are eye-catching linkers with an extensive assortment of applications. In the synthesis of bioactive and functional molecules, linking two pharmacophores to create innovative bifunctional medications has grown more practical and vital.⁶²⁻⁶⁷ Sulfonamides, also known as sulfa medicines, are among the oldest pharmaceutical agents used as preventive and chemotherapeutic agents against a variety of medical conditions.^{68,69} Hybridization of all three moieties in one scaffold could produce derivatives with synergistic antibacterial effects. A series of novel benzimidazole-triazole-sulfonamide hybrids were synthesized (Fig.-15) and recorded for antibacterial activity against tested bacterial strains. MIC values of compounds showed that compound 6a was reported to be a potent antibacterial compound (MIC 32-64 µg/mL). Mono derivatives of 1, 2, and 3-triazole (45a-f) displayed a reduction of antibacterial activity in contrast to bis derivatives (46a-f).⁷⁰

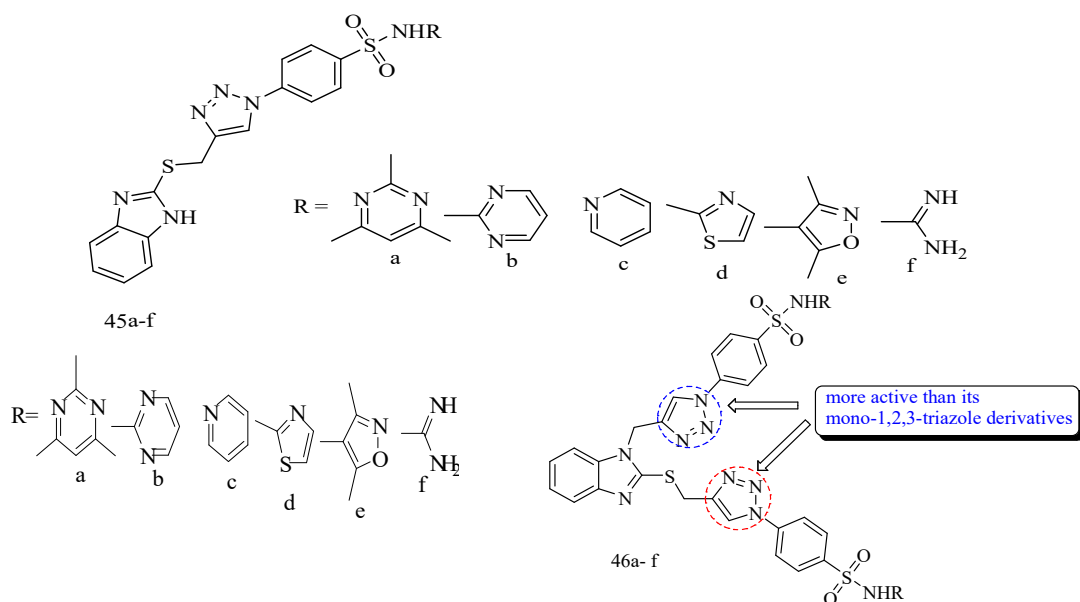


Fig.-15: Benzimidazole1-Sulfonamide Hybrids

Antibacterial Activity of New Quinazoline-Benzimidazole Hybrid Derivatives

Quinazoline is a unique moiety with remarkable activities such as antitubercular, antibacterial (Fig.-16)⁷¹, anticonvulsant⁷², antifungal⁷³, anti-inflammatory⁷⁴, and anti-tumor.⁷⁵ 4(3H)-quinazolinones have been identified as an orally accessible antibacterial agent by Bouley *et al.*⁷⁶ Antibacterial and antimycobacterial properties of 15 novel quinazoline benzimidazole synthesized hybrids were assessed. Findings indicated that the synthesized compounds had *S. aureus*-specific inhibitory action. With a MIC of 4 µg/mL, Derivatives 47b, 47c, 47h, and 47i (Fig.-17) showed similar antibacterial action. Derivatives 47g, 48a, 48b, and 49 showed considerable potency with MICs of 16 to 64 µg/mL. The alteration in -COOH group at the C-4 phenyl (at the para position) by coupling with modified benzimidazoloamines to different amides as a result, the derivatives 48a, 48b, 48c, and 48d have been created. With MIC (32- 64 µg/mL), the synthesized compounds displayed mild suppression against *S. aureus*. At C-4, 2-trifluoromethyl benzimidazolo-5-amine was substituted for 2-trifluoromethyl benzimidazolo-5-amine, yielding compound 49 and discovered to be inert. Compound 47f, with a MIC of 8 µg/mL, was found to have strong inhibitory efficacy against *M. tuberculosis*.⁷⁷

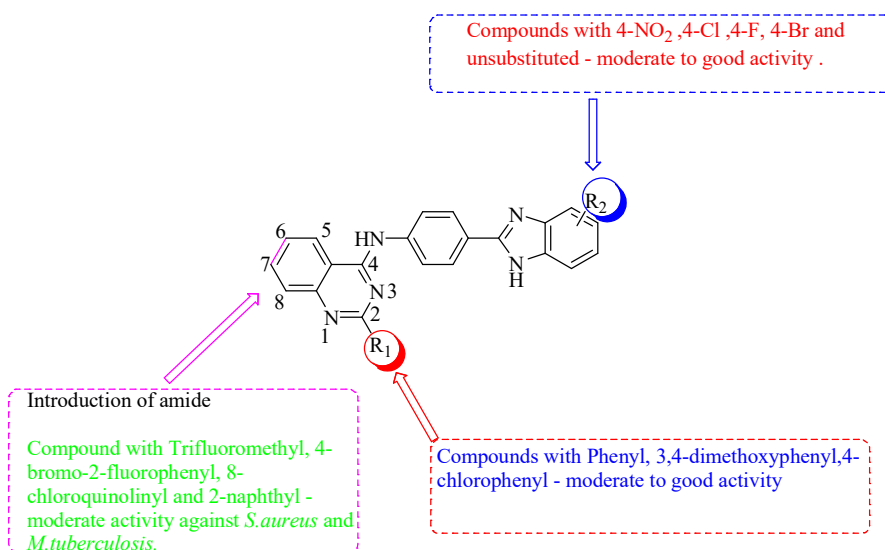


Fig.-16: Antimicrobial Action Requires Structural Constraints Around the Benzimidazole- Quinazoline Nucleus

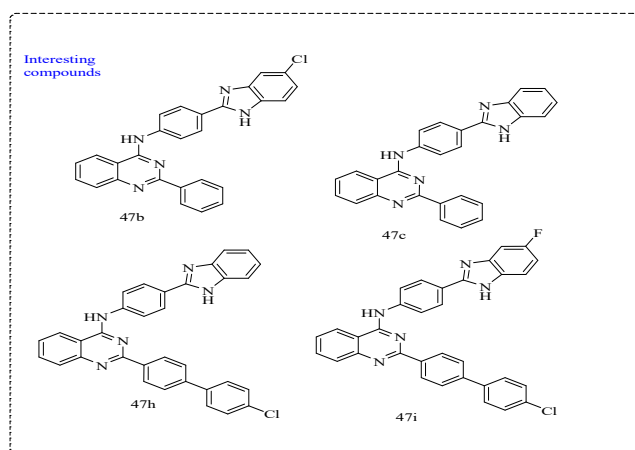


Fig.-17: Structures of Quinazoline-Benzimidazole Hybrid Derivatives found to be potent antibacterial agents such as: 47b, 47c, 47h, and 47i

CONCLUSION

Antibiotic resistance is predicted to kill ten million people each year by 2050, in accordance with the World Health Organization, necessitating the availability of novel medicines. Benzimidazole derivatives

are crucial inhibitors of DNA gyrase, Pyruvate kinase, Topoisomerase IV, GTPase, NDM1 enzyme, and efflux pump indicating that they could be used in mitigation of various pathogenic infections. Hybridization is a popular method in current drug development, combining the benzimidazole moiety with other antibacterial pharmacophores could boost antibacterial efficacy, and may produce the novel antibacterial agents. Based on the findings, it can be inferred that hybrids with electronegative groups for example- Chloro, Hydroxyl, Methoxy, Amino methyl, Carboxyl methyl, Amide, Aldehyde, Trifluoromethyl, Nitro, Cyano, Halogens attached on the both ring benzimidazole as well as other heterocyclic nucleus such as thiazole, indole, pyridine, pyrimidine further improve the antibacterial potential of the compounds. Furthermore, if the compound has linkers such as amide (NHCO), carbonyl (CO) or other hetero atoms, the antibacterial activity is enhanced. We believe that this work could be a springboard to develop hybrids of benzimidazole with other heterocyclic as novel antibacterial agents.

ACKNOWLEDGMENTS

We would like to thank all of the co-authors for their insightful criticism and assessment of the paper. Manuscript Communication Number (MCN): IU/ R&D/ 2022-MCN0001512 Integral University, Lucknow, R&D/office of research and development.

CONFLICT OF INTERESTS

The authors declare that there is no conflict of interest.

AUTHOR CONTRIBUTIONS

All the authors contributed significantly to this manuscript, participated in reviewing/editing, and approved the final draft for publication. The research profile of the authors can be verified from their ORCID ids, given below:

Suvaiv  <https://orcid.org/0000-0003-0240-8029>

Kuldeep Singh  <https://orcid.org/0000-0001-5418-945X>

Praveen Kumar  <https://orcid.org/0000-0003-1091-6840>

Syed Misbahul Hasan  <https://orcid.org/0000-0003-2189-2688>

Shom Prakash Kushwaha  <https://orcid.org/0000-0002-9844-400X>

Arun Kumar  <https://orcid.org/0000-0002-6323-7377>

Syed Mehdi Hasan Zaidi  <https://orcid.org/0009-0001-5496-6316>

Shubham Kumar Maurya  <https://orcid.org/0009-0007-7802-2954>

Open Access: This article is distributed under the terms of the Creative Commons Attribution 4.0 International License (<http://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

REFERENCES

1. M. Huemer, S. M. Shambat, S. D. Brugger, A. S. Zinkernagel, *European Molecular Biology Organization Reports*, **21**(12), 51034 (2020), <https://doi.org/10.15252/embr.202051034>
2. M. Giannella, M. Bartoletti, M. Gatti, P. Viale, *Clinical Microbial Infection*, **26**(2), 158 (2020), <https://doi.org/10.1016/j.cmi.2019.11.001>
3. G. Dhanda, P. Sarkar, S. Samaddar, J. Halder, *Journal of Medicinal Chemistry*, **62**(7), 3184 (2018), <https://doi.org/10.1021/acs.jmedchem.8b01093>
4. M. Liu, E. M. El-Hossary, T. A. Oelschlaeger, M. S. Donia, R. J. Quinn, U.R. Abdelmohsen, *Lancet Infectious Diseases*, **19**(7), 237 (2019), [https://doi.org/10.1016/S1473-3099\(18\)30711-4](https://doi.org/10.1016/S1473-3099(18)30711-4)
5. C. Gao, Y. L. Fan, Zhao, F., Ren, Q. C., Wu, X., Chang, L., & Gao, F. *European Journal of Medicinal Chemistry*, **157**, 1081 (2018), <https://doi.org/10.1016/j.ejmech.2018.08.061>
6. F. Gao, J. Xiao, G. Huang, *European Journal of Medicinal Chemistry*, **184**, 1117 (2019), <https://doi.org/10.1016/j.ejmech.2019.111744>

7. C. Painter, D. Faradiba, K. K. Chavarina, E. N. Sari, Y. Teerawattananon, K. Aluzaita, Ananthakrishnan *Antimicrobial Resistance and Infection Control*, **12(1)**, 1(2023), <https://doi.org/10.1186/s13756-023-01265-5>
8. <https://www.health.europa.com/gram-report-antimicrobial-resistance-involved-in-five-million-deaths/112908/>.
9. M. Garland, S. Loscher, M. Bogyo, *Chemical Review*, **117(5)**, 4422(2017), <https://doi.org/10.1021/acs.chemrev.6b00676>
10. B. Gadhiya, M. Rajput, A. Bapodra, K. Ladva, *Rasayan Journal of Chemistry*, **9(3)**, 355(2016).
11. R. Kenchappa, Y. D. Bodke, S. Telkar, and M. Aruna Sindhe, *Journal of chemical Biology*, **10(1)**, 11(2016), <https://doi.org/10.1007/s12154-016-0160-x>
12. A. T. Taher, N. A. Khalil, E. M. Ahmed, *Archives of Pharmacal Research*, **10(1)**, 1615(2011), <https://doi.org/10.1007/s12272-011-1005-3>
13. P. Kagthara, T. Upadhyay, R. Doshi, H. H. Parekh, *Indian Journal of Heterocyclic Chemistry*, **13(3)**, 9(2000).
14. A. K. Bhatt, H. G. Karadia, P. R. Shah, M. P. Parmar, H. D. Patel, *Indian Journal of Heterocyclic Chemistry*, **14**, 281(2004)
15. V. K. Pandey, V. D. Gupta, D. N. Tiwari, *Indian Journal of Heterocyclic Chemistry*, **2(3)**, 217(2005).
16. G. S. Babu, N. Rajani, P. S. Malathy, B. Srinivas, U. Kulandaivelu, J. V. Rao, *Der Pharma Chemica*, **34(10)**, 196(2010).
17. B. Baviskar, S. Chuadhary, K. Parwani, P. Balani, V. Salode, S. Patil, S. S. Khadabadi, *Rasayan Journal of Chemistry*, **2(1)**, 186(2009).
18. K. Parikh, D. Joshi, *Medicinal Chemistry Research*, **22(8)**, 3688(2013), <https://doi.org/10.1007/s00044-012-0369-3>
19. E. Menteşe, H. Bektaş, B. B. Sokmen, M. Emirik, D. Çakır, B. Kahveci, *Bioorganic & Medicinal Chemistry Letters*, **27(13)**, 3014(2017), <https://doi.org/10.1016/j.bmcl.2017.05.019>
20. F. Yılmaz, E. Menteşe, N. Baltas, *Letters in Drug Design and Discovery*, **14(2)**, 201(2017), <https://doi.org/10.2174/1570180813666160609082633>
21. O. Bekircan, N. Baltas, E. Menteşe, E. Gultekin, *Romanian Journal of Chemistry*, **61(10)**, 733(2016).
22. E. Menteşe, N. Karaali, G. Akyüz, F. Yılmaz, S. Ülker, B. Kahveci, *Chemistry of Heterocyclic Compound*, **52(12)**, 1017(2016), <https://doi.org/10.1007/s10593-017-2002-3>
23. B. Kahveci, F. Yılmaz, E. Menteşe, S. Ülker, *Archives of Pharmacal Research*, **350(8)**, 1600369(2017), <https://doi.org/10.1002/ardp.201600369>
24. F. Yılmaz, E. Menteşe, *Journal of Chemical Research*, **41(1)**, 4(2017), <https://doi.org/10.3184/174751917X148154272190>
25. N. Baltas, F. YILMAZ, E. Menteşe, *Hacettepe Journal of Biology and Chemistry*, **44(3)**, 293(2016), <https://doi.org/10.15671/HJBC.20164420572>
26. M. Emirik, K. Karaoglu, K. Serbest, E. Menteşe, I. Yilmaz, *Journal of Molecular Structure*, **1106**, 331(2016), <https://doi.org/10.1016/j.molstruc.2015.10.075>
27. E. Menteşe, B. Kahveci, *Chemistry of Heterocyclic Compounds*, **52(11)**, 948(2016), <https://doi.org/10.1007/s10593-017-1991-2>
28. I. S. Krishnanjaneyulu, G. Saravanan, J. Vamsi, P. Supriya, J. U. Bhavana, M. V. Kumar, *Journal of Advanced Pharmaceutical Technology & Research*, **5(1)**, 21(2014), <https://doi.org/10.4103/2231-4040.126983>
29. N. C. Desai, G. M. Kotadiya, *Medicinal Chemistry Research*, **23(9)**, 4021(2014), <https://doi.org/10.1007/s00044-014-0978-0>
30. D. Sharma, B. Narasimhan, P. Kumar, V. Judge, R. Narang, E. De Clercq and J. Balzarini, *European Journal of Medicinal Chemistry*, **44(6)**, 2347(2009), <https://doi.org/10.1016/j.ejmech.2008.08.010>
31. N. C. Desai, D. D. Pandya, G. M. Kotadiya, P. Desai, *Medicinal Chemistry Research*, **23(3)**, 1474(2014), <https://doi.org/10.1007/s00044-013-0756-4>
32. N. C. Desai, N. R. Shihory, G. M. Kotadiya, *Chinese Chemical Letters*, **25(2)**, 305(2014).

33. P. Picconi, C. Hind, S. Jamshidi, K. Nahar, M. Clifford, M. E. Wand, J. M. Sutton and K. M. Rahman, *Journal Medicinal Chemistry*, **60(14)**, 6045(2017), <https://doi.org/10.1021/acs.jmedchem.7b00108>
34. G. Ayhan-Kılıçgil, N. Altanlar, *Il Farmaco*, **58(12)**, 1345(2003), [https://doi.org/10.1016/S0014-827X\(03\)00190-3](https://doi.org/10.1016/S0014-827X(03)00190-3)
35. A. Khalafi-Nezhad, M. S. Rad, H. Mohabatkar, Z. Asrari, B. Hemmateenejad, *Bioorganic & Medicinal Chemistry*, **13(6)**, 1931(2005), <https://doi.org/10.1016/j.bmc.2005.01.014>
36. L. R. Singh, S. R. Avula, S. Raj, A. Srivastava, G. R. Palnati, C. K. M. Tripathi, M. Pasupuleti and K. V. Sashidhara, *The Journal of Antibiotics*, **70(9)**, 954(2017), <https://doi.org/10.1038/ja.2017.70>
37. N. Shruthi, B. Poojary, V. Kumar, M. M. Hussain, V. M. Rai, V. R. Pai, M. Bhat, B. C. Revannasiddappa, *RSC advances*, **6(10)**, 8303(2016), <https://doi.org/10.1039/C5RA23282A>
38. T. L. S. Kishbaugh, *Current Topics in Medicinal Chemistry*, **16(28)**, 3274(2016), <https://doi.org/10.2174/1568026616666160506145141>
39. H. A. Dansena, H. J. Dhongade, K. A. Chandrakar, *Asian Journal of Pharmaceutical and Clinical Research*, **8(4)**, 171(2015).
40. S. Cherukupalli, R. Karpoormath, B. Chandrasekaran, G. A. Hampannavar, N. Thapliyal, V. N. Palakollu, *European Journal of Medicinal Chemistry*, **126**, 298(2017), <https://doi.org/10.1016/j.ejmech.2016.11.019>
41. N. C. Desai, D. D. Pandya, V. V. Joshi, K. M. Rajpara, H. V. Vaghani, H. M. Satodiya, *Medicinal Chemistry Research*, **21(12)**, 4463(2012), <https://doi.org/10.1007/s00044-012-9990-4>
42. X. Song, B. S. Vig, P. L. Lorenzi, D. C. Drach, L. B. Townsend, G. L. Amidon, *Journal of Medicinal Chemistry*, **48(4)**, 1274(2005), <https://doi.org/10.1021/jm049450i>
43. D. Kumar, M. R. Jacob, M. B. Reynolds, S. M. Kerwin, *Bioorganic and Medicinal Chemistry*, **10(12)**, 3997(2002), [https://doi.org/10.1016/s0968-0896\(02\)00327-9](https://doi.org/10.1016/s0968-0896(02)00327-9)
44. I. Yildiz-Oren, I. Yalcin, E. Aki-Sener, N. Ucarturk, *European Journal of Medicinal Chemistry*, **39(3)**, 291(2004), <https://doi.org/10.1016/j.ejmech.2003.11.014>
45. G. J. Reddy, D. Latha, C. Thirupathaiah, K. S. Rao, *Heterocyclic Communications*, **10(4-5)**, 359(2004), <https://doi.org/10.1515/HC.2004.10.4-5.359>
46. F. E. Boyer, J. V. Vara Prasad, A. L. Choy, L. Chupak, M. R. Dermeyer, Q. Ding, M. D. Huband, W. Jiao, T. Kaneko, V. Khlebnikov, J. Y. Kim, M. S. Lall, S. N. Maiti, K. Romero and X. Wu, *Bioorganic and Medicinal Chemistry Letters*, **17(16)**, 4694(2007), <https://doi.org/10.1016/j.bmcl.2007.05.056>
47. V. S. Padalkar, B. N. Borse, V. D. Gupta, K. R. Phatangare, V. S. Patil, N. Sekar, *Journal of Heterocyclic Chemistry*, **53(5)**, 1347(2016), <https://doi.org/10.1002/jhet.1506>
48. N. Harikrishna, A. M. Isloor, K. Ananda, A. Obaid, H. K. Fun, *New Journal of Chemistry*, **40(1)**, 73(2016), <https://doi.org/10.1039/C5NJ02237A>
49. S. D. Joshi, S. R. Dixit, M. N. Kirankumar, T. M. Aminabhavi, K. V. Raju, R. Narayan, C. Lherbet, and K. S. Yang, *European Journal of Medicinal Chemistry*, **107**, 133(2016), <https://doi.org/10.1016/j.ejmech.2015.10.047>
50. P. Ahmad, H. Woo, K. Y. Jun, A. A. Kadi, H. A. Abdel-Aziz, Y. Kwon and A. F. Rahman, *Bioorganic & Medicinal Chemistry*, **24(8)**, 1898(2016), <https://doi.org/10.1016/j.bmc.2016.03.017>
51. Y. J. Qin, Y. J. Li, A. Q. Jiang, M. R. Yang, Q. Z. Zhu, H. Dong and H. L. Zhu, *European Journal of Medicinal Chemistry*, **94**, 447(2015), <https://doi.org/10.1016/j.ejmech.2015.02.058>
52. K. R. Abdellatif, E. K. Abdelall, W. A. Fadaly, G. M. Kamel, *Bioorganic and Medicinal Chemistry Letters*, **26(2)**, 406(2016), <https://doi.org/10.1016/j.bmcl.2015.11.105>
53. E. K. Abdelall, P. F. Lamie, W. A. Ali, *Bioorganic and Medicinal Chemistry Letters*, **26(12)**, 2893(2016), <https://doi.org/10.1016/j.bmcl.2016.04.046>
54. M. Jagrat, J. Behera, S. Yabanoglu, A. Ercan, G. Ucar, B. N. Sinha, V. Sankaran, A. Basu, and V. Jayaprakash, *Bioorganic & Medicinal Chemistry Letters*, **21(14)**, 4296(2011), <https://doi.org/10.1016/j.bmcl.2011.05.057>
55. B. Evranos-Aksöz, S. Yabanoğlu-Çiftçi, G. Uçar, K. Yelekçi, R. Ertan, *Bioorganic and Medicinal Chemistry Letters*, **24(15)**, 3278(2014), <https://doi.org/10.1016/j.bmcl.2014.06.015>

56. S. S. Sulthana, S. A. Antony, C. Balachandran, S. S. Shafi, *Bioorganic and Medicinal Chemistry Letters*, **25(14)**, 2753(2015), <https://doi.org/10.1016/j.bmcl.2015.05.033>
57. G. K. Padhy, J. Panda, A. K. Behera, *arXiv*, **2005**, 01375(2022), <https://doi.org/10.48550/arXiv.2005.01375>
58. K. V. Chikkula, R. Sundararajan, *Medicinal Chemistry Research*, **26(11)**, 3026(2017), <https://doi.org/10.1007/s00044-017-2000-0>
59. M. Mostafa, Abdelgalil, A. Yousry, Ammar, A. M. Gameel, Elhag Ali, K. H. Ali, A. Ragab. *Journal of Heterocyclic Chemistry*, **56(2)**, 589(2019), <https://doi.org/10.1002/jhet.3435>
60. K. Chkirate, K. Karrouchi, N. Dege, N. K. Sebbar, A. Ejjoummany, S. Radi, N. N. Adarsh, A. Talbaoui, M. Ferbinteanu, E. M. Essassi, Y. Garcia, *New Journal of Chemistry*, **44(6)**, 2210(2020), <https://doi.org/10.1039/C9NJ05913J>
61. F. A. Bassyouni, H. A. Tawfik, A. R. Hamed, M. M. Soltan, M. ElHefnawi, A. A. ElRashedy, M. E. Moharam, M. A. Rehim, *Egyptian Pharmaceutical Journal*, **11(2)**, 80(2012), <https://doi.org/10.7123/01.EPJ.0000422113.69898.e0>
62. R.Kant, V. Singh, G. Nath, S. K. Awasthi, A. Agarwal, *European Journal of Medicinal Chemistry*, **29**, 218(2016), <https://doi.org/10.1016/j.ejmech.2016.08.031>
63. T. Gregorić, M. Sedić, P. Grbčić, A. T. Paravić, S. K. Pavelić, M. Cetina, R. Vianello, S. Raić-Malić, *European Journal of Medicinal Chemistry*, **125**, 1247(2017), <https://doi.org/10.1016/j.ejmech.2016.11.028>
64. A. Ouahrouch, H. Ighachane, M. Taourirte, J. W. Engels, M. H. Sedra, H. B. Lazrek, *Archives of Pharmacol Research*, **347(10)**, 748(2014), <https://doi.org/10.1002/ardp.201400142>
65. Z. Xu, S. Zhang, X. Song, M. Qiang, Z. Lv, *Bioorganic and Medicinal Chemistry Letters*, **27(16)**, 3643(2017), <https://doi.org/10.1016/j.bmcl.2017.07.023>
66. S. Sharma, M. Saquib, S. Verma, N. N. Mishra, P. K. Shukla, R. Srivastava, Y. S. Prabhakar and A. K. Shaw, *European Journal of Medicinal Chemistry*, **83**, 474(2014), <https://doi.org/10.1016/j.ejmech.2014.06.048>
67. X. Yan, Z. Lv, J. Wen, S. Zhao, Z. Xu, *European Journal of Medicinal Chemistry*, **143**, 899(2018), <https://doi.org/10.1016/j.ejmech.2017.11.090>
68. C. T. Supuran, *Molecules*, **22(10)**, 1642(2017), <https://doi.org/10.3390/molecules22101642>
69. A. Ajeet, A. K. Mishra, A. Kumar, *American Journal of Pharmacology*, **3(1)**, 18(2015), <https://doi.org/10.12691/ajps-3-1-4>
70. F. F. Al-Blewi, M. A. Almeahadi, M. R. Aouad, S. K. Bardaweel, P. K. Sahu, M. Messali, N. Rezki and E. S. H. El Ashry, *Chemistry Central Journal*, **12(1)**, 1(2018), <https://doi.org/10.1186/s13065-018-0479-1>
71. J. Kuneš, J. Bažant, M. Pour, K. Waisser, M. Šlosárek, J. Janota, *Il Farmaco*, **55(11-12)**, 725(2020), [https://doi.org/10.1016/s0014-827x\(00\)00100-2](https://doi.org/10.1016/s0014-827x(00)00100-2)
72. V. G. Ugale, S. B. Bari, *European Journal of Medicinal Chemistry*, **80**, 447(2014), <https://doi.org/10.1016/j.ejmech.2014.04.072>
73. F. Hassanzadeh, E. Jafari, G. H. Hakimelahi, M. R. Khajouei, M. Jalali, G. A. Khodarahmi, *Research in Pharmaceutical Sciences*, **7(2)**, 87(2012).
74. D. B. Farag, N. A. Farag, A. Esmat, S. A. Abuelezz, E. A. Ibrahim, D. A. Abou El Ella, *Medicinal Chemistry Communications*, **6(2)**, 283(2015), <https://doi.org/10.1039/C4MD00392F>
75. M. K. Krapf, J. Gallus and M. Wiese, *Journal of Medicinal Chemistry*, **60(10)**, 4474(2017), <https://doi.org/10.1021/acs.jmedchem.7b0044>
76. R. Bouley, D. Ding, Z. Peng, M. Bastian, E. Lastochkin, W. Song, M. A. Suckow, V. A. Schroeder, W. R. Wolter, S. Mobashery and M. Chang, *Journal of Medicinal Chemistry*, **59(10)**, 5011(2016), <https://doi.org/10.1021/acs.jmedchem.6b00372>
77. S. Malasala, M. N. Ahmad, R. Akunuri, M. Shukla, G. Kaul, A. Dasgupta, Y. V. Madhavi, S. Chopra, and S. Nanduri, *European Journal of Medicinal Chemistry*, **212**, 112996(2021), <https://doi.org/10.1016/j.ejmech.2020.112996>

[RJC- 8382/2023]